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Digital Dentistry in Operative Dentistry

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Digital technology, computerized dentistry, and digital dentistry are general terms used to describe the clinical application of computer-assisted design, computer-assisted machining (CAD/CAM). The restorative dentistry application of CAD/CAM technology is the fabrication and delivery of permanent restorations for teeth and implants. For the past 30 years the incorporation of dental CAD/CAM into direct patient care has provided a way for dentists to deliver esthetic ceramic restorations in a single dental appointment.

There are three sequences involved in the CAD/CAM process.[1] An intraoral scanner or camera is used to accurately record the hard and soft tissue geometry of the patient's intraoral condition to a computer program in the first sequence. This is commonly referred to as a digital impression. A proprietary software design program is used to create a virtual restoration (the volume proposal) in the second sequence. The software programs have the capability of controlling and editing the various parameters of the restoration such as emergence profile, proximal contact, and occlusal relationships. Once the proposal of the restoration has been completed, a computer-controlled device is used to produce the restoration in the third sequence. The most common device uses a subtractive process to machine (i.e., grind or mill, depending on whether carbide burs or diamonds are used) the final restoration from a preformed block of a variety of restorative materials. Understanding these three sequences provides a basis for deciding how best to implement the technology in the dental office and also creates a simple way to categorize various systems in the marketplace.

Digital impression systems are designed to accurately record the intraoral geometry and then transmit the files to a dental laboratory for design and fabrication of the desired restoration. There is minimal if any opportunity for the dentist to design any aspect of the restoration on digital impression systems. They were developed to leverage the digital recording process to take advantage of the comfort and efficiency of not using conventional impression materials as well as the convenience and accuracy of digital transmission of the case to the dental laboratory.

Chairside CAD/CAM systems employ all three sequences of the CAD/CAM process in the dental office. They also record intraoral scans but provide, in addition, a software program for designing restorations as well as milling units to fabricate the restoration during a single dental appointment. They are designed to leverage the efficiency of a single appointment procedure for the delivery of ceramic restorations. Chairside CAD/CAM systems are the primary focus of the chapter because they allow the dentist complete control of the design, fabrication, and delivery of the restoration in the office during a single appointment.

Clinical Application

Treatment planning considerations for CAD/CAM restorations are not significantly different from ceramic restorations done with conventional impression materials and techniques. The CAD/CAM system represents an alternative means of restoration fabrication, not the restoration per se. The type of restoration (inlay, onlay, crown), the choice of material to be used, the desired occlusal relationships, and ability to isolate the tooth preparation for delivery of the restoration are several primary factors to consider rather than the restoration fabrication process itself. A case in point is that the predictable ability to isolate a subgingival margin for adhesive cementation is a much more important factor to consider than whether the ceramic restoration is fabricated with a conventional or digital impression technique. Nonetheless, there are a few specific considerations relative to the use of digital impressions. The relative size of the camera may be a concern for patients with a restricted ability to open wide. Generally, if there is sufficient vertical space to complete the tooth preparation with a dental handpiece, there is sufficient space for use of a digital camera. However, patients with a severe gag reflex may appreciate the use of a digital impression more than conventional impression since there is no physical contact with the intraoral tissues by a tray or impression material when recording a digital impression.

Chairside CAD/CAM Systems

Dr. Francois Duret conceptualized the first chairside CAD/CAM system in 1973. The first functioning chairside CAD/CAM prototype was introduced in the 1980s through the collaboration between a Swiss prosthodontist, Dr. Werner Mörmann, and an Italian electrical engineer, Marco Brandestini. Dr. Mörmann's vision was to use CAD/CAM technology to deliver esthetic ceramic restorations with improved longevity in a single appointment that avoided the deleterious consequences polymerization shrinkage caused in composite restorations.[2,3] The CEREC 1 unit marked the introduction of the CEREC system in 1985 with the first clinical trials reported in 1987.[4] The system has evolved through a series of hardware and software innovations and upgrades that culminated with the introduction of the first color-streaming powder-free intraoral camera in 2012, the CEREC Omnicam (Fig. 12.1).[5,6] The Omnicam is the imaging camera of the CEREC AC acquisition unit. The Omnicam is connected to a chairside computer with the three-dimensional design software and a liquid crystal display (LCD) monitor. The CEREC system is electronically connected to a milling unit. The newest version is a dry grinding



• **Fig. 12.1** CEREC OmniCam and MCX milling chamber. (Courtesy Dentsply Sirona.)



• **Fig. 12.2** PlanFit chairside CAD/CAM system. (Courtesy D4D Technologies.)

and wet milling unit that is the first in-office means by which to dry mill zirconia. Single-visit, full-contour, milled zirconia restorations are then further processed by sintering and glazing the restoration in the CEREC SpeedFire sintering furnace. This is the first in-office zirconia sintering furnace that can sinter preshaded CEREC zirconia material (CEREC Zirconia) in 10 to 15 minutes, as compared to the multiple hours that are generally necessary for sintering laboratory processed zirconia restorations.

An increasing number of chairside systems have been introduced since the late 2000s. The E4D Dentist System (D4D Technologies) was introduced in 2008 with its DentaLogic software offering a true three-dimensional virtual model.[7] The E4D Dentist System has an intraoral laser scanner, mobile Design Center with DentaLogic software, and a separate milling unit with a dedicated CAM server computer. The milling unit has two opposing electric motors that automatically change between three different diamonds depending on the specifics of the restoration dimensions. The milling unit has a dedicated computer server, which allows independent operation separate from the Design Center (after completion and transmission of the case design) (Fig. 12.2).

The CS 3500 is a powder-free intraoral scanner that was introduced in 2013. The Carestream system allows for in-office design using CS Solutions Restore software and in-office fabrication using the CS 3000 milling unit. The digital files may also be

uploaded to a dental laboratory via CS Connect as they are compatible with a number of commercial laboratory design programs such as 3Shape (3Shape) and Exocad (Exocad Gmbh). Some of the unique features of the intraoral scanner include a disposable tip and a guiding light, which indicates a successful scan. The Carestream system has chairside capability for only a limited number of applications, including inlays, onlays, and crowns.

Chairside CAD/CAM systems are able to produce inlays, onlays, veneers, and crowns. Some systems also have the ability to fabricate short-span fixed partial dentures as well as temporary restorations in the dental office. Additional applications, such as implant abutments, fixed partial dentures, and orthodontic appliances, are unique to specific systems (Table 12.1).

Tooth Preparation Principles for CAD/CAM Restorations

Guidelines for tooth preparation for all-ceramic restorations are generally based on the specific geometries and thickness dimensions required to provide optimum strength for the selected ceramic material. For example, the occlusal reduction for a full-contour zirconia crown is less than that required for a leucite-reinforced or feldspathic porcelain crown, even if both materials are fabricated with a chairside CAD/CAM system. However, there are several preparation guidelines that enhance the accuracy of the CAD/CAM restoration due to their influence on the imaging and fabrication process, such as smooth contours, rounded transitions, and uniform pulpal floor.

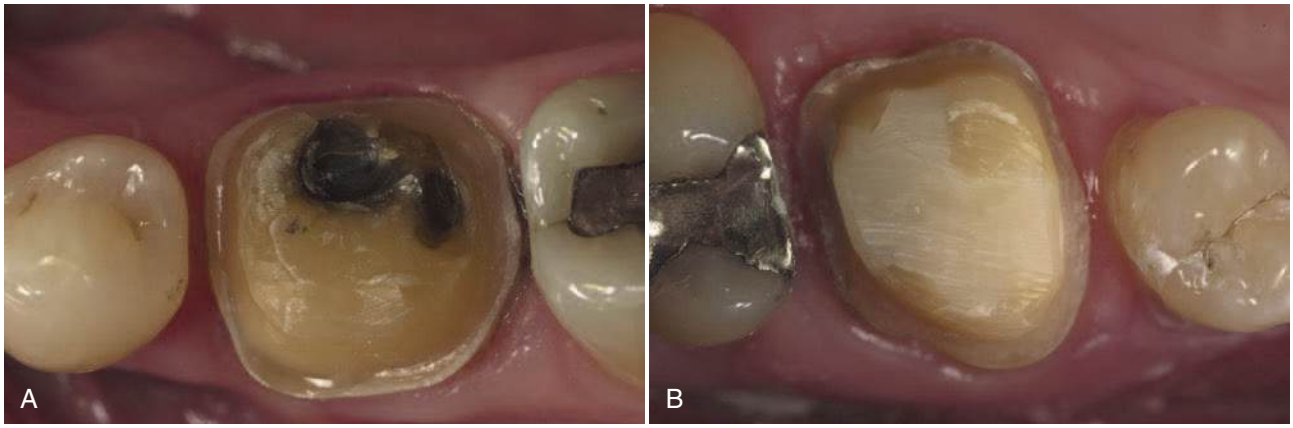
A concept unique to CAD/CAM restorations has been termed "undermilling" or "overmilling." These terms have been applied to both grinding and milling processes. Any preparation geometry smaller than the dimension of the milling instruments presents a potential problem. Either the instrument must cut away more restorative material to ensure the restoration seats completely (overmilling) or it does not remove the smaller geometry (undermilling), leaving restorative material that will prevent complete seating of the restoration. Utilization of a preparation geometry with smooth contours, gently flowing curves, and rounded transition angles helps to limit the potential for overmilling and ensure an accurate internal adaptation of the restoration.

Crowns

Tooth preparation for CAD/CAM-fabricated crowns is essentially the same as for laboratory-fabricated ceramic crowns. The margin design for an all-ceramic crown requires a bulk of ceramic at the margin to avoid the risk of ceramic chipping or fracture. This is best accomplished with a shoulder, sloped shoulder, or heavy chamfer margin (Fig. 12.3). All of these margin geometries are equally useful for chairside CAD/CAM crowns as long as the internal angles are rounded (i.e., not sharp or angular) to facilitate internal adaptation of the crown. The occlusal and axial reduction requirements are determined based on the specific material properties of the ceramic material selected for the crown. Axial reduction for CAD/CAM ceramic crowns should be 1 to 1.2 mm for the monolithic ceramic materials as there is no opaque inner coping that needs to be covered (veneered). A greater variation in occlusal

TABLE 12.1 Clinical Applications for Chairside CAD/CAM Systems

System	Inlays/Onlays	Crowns	Veneers	FPDs	Implant Abutment	Orthodontic Application	CBCT Compatible
CEREC (Dentsply Sirona)							
BlueCam	X	X	X	X	X	X	X
OmniCam	X	X	X	X	X	X	X
PlanFit (Planmeca)	X	X	X	X	–	–	X
CS Solutions (Carestream)	X	X	X	–	–	–	X



• **Fig. 12.3** A and B, Examples of chairside CAD/CAM crown preparations.



• **Fig. 12.4** A and B, Examples of the occlusal clearance for chairside CAD/CAM crown preparations.

reduction may be expected as a function of the material selected. Most glass ceramic materials require a minimum of 1.5 mm occlusal reduction in the central fissure area and over nonfunctional cusps and 2 mm over functional cusps (Fig. 12.4). Laboratory-based (*in vitro*) research studies suggest that full-contour zirconia crowns may be fabricated with an occlusal thickness of 1 mm without compromising the strength of the crown. Additionally, the manufacturer of lithium disilicate blocks has recently announced that e.maxCAD crowns now only require a 1-mm occlusal reduction; however, this claim still requires independent validation. All angles of the preparation should be rounded to facilitate accurate internal adaptation and needless overcutting of the internal surface of the crowns during milling.

It is not uncommon to identify significant undercuts in a crown preparation as a result of prior large restorations or caries excavation. Undercuts will be accurately recorded by the digital camera and be present in the calculated virtual model in the CAD software program. However, the milling process will not duplicate any internal undercuts in the preparation that might prevent seating of the restoration. These undercuts would subsequently be filled in with the resin cement. Alternatively, undercuts could be blocked out using the software program during design of the restoration. This process is not recommended, however, as it could ultimately result in an internal inaccuracy as the digital model is altered. Therefore, if blockout is necessary, it is best accomplished with an appropriate dentin substitute material, as indicated, prior to digital scanning.



• **Fig. 12.5** Onlay preparations for teeth #3 and #4 relying primarily on the adhesive bond of the resin cement for retention of the restoration.

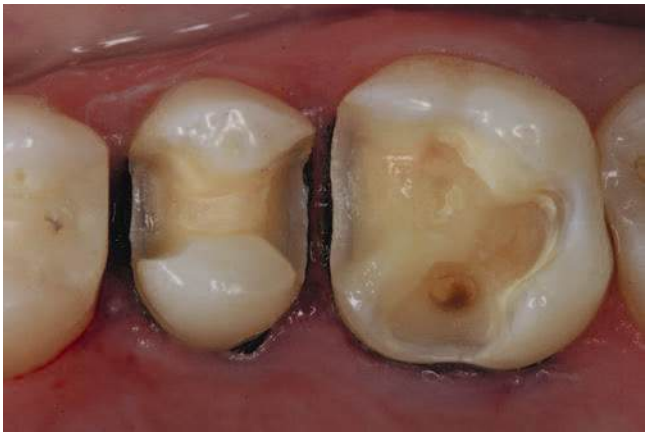


• **Fig. 12.6** Full cuspal onlay preparation for tooth #3 and MODL onlay for #4 with smooth and rounded internal angles.

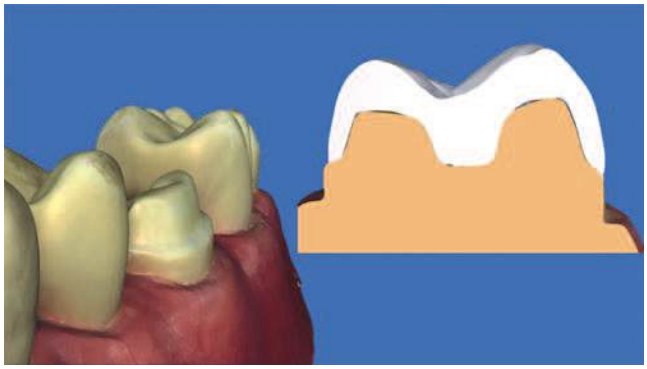
Inlays and Onlays

CAD/CAM inlay and onlay preparations are primarily adhesive-style preparations that rely on the adhesion of the resin cement to dentin and enamel for retention of the restoration (Fig. 12.5). These preparations are divergent and relatively nonmechanically retentive in design as this provides a more conservative preparation than the requirement for mechanical resistance through grooves, slots, or boxes. The internal aspect of the preparation should avoid sharp divots or concavities, and all internal angles should be rounded (Fig. 12.6). Occlusal reduction should be uniform and of sufficient thickness to provide optimum strength of the selected ceramic material similar to crown preparations. Preparation should allow for a minimum of 1.5 mm of ceramic thickness in the central fossa and over nonfunctional cusps, and 2 mm over functional cusps. All cavosurface margins should be strategically placed away from the contact position of the opposing cusp(s) and be well defined (smooth) for easy identification in the design software. Beveled margins must be avoided, as thin areas of ceramic are prone to fracture.[8,9] Similarly, the preparation isthmus should be at least 2 mm in faciolingual width so as to avoid inlay/onlay fracture (Fig. 12.7).

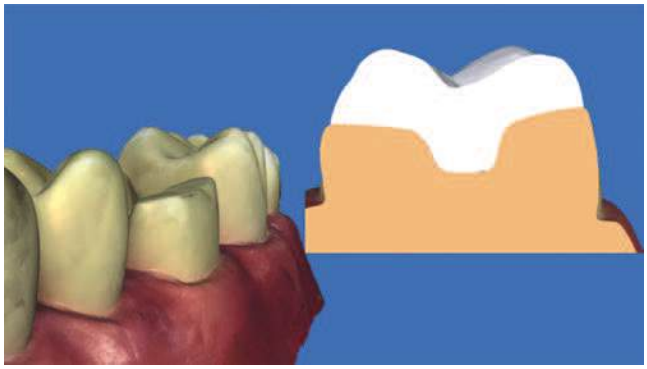
CAD/CAM onlay preparations do not require the creation of a ferrule with the ceramic restoration as used with metal castings (Fig. 12.8). On the contrary, a ferrule may actually interfere with



• **Fig. 12.7** Inlay preparation for tooth #13 and onlay preparation for tooth #14 with smooth flowing outline form and crisp cavosurface margins.

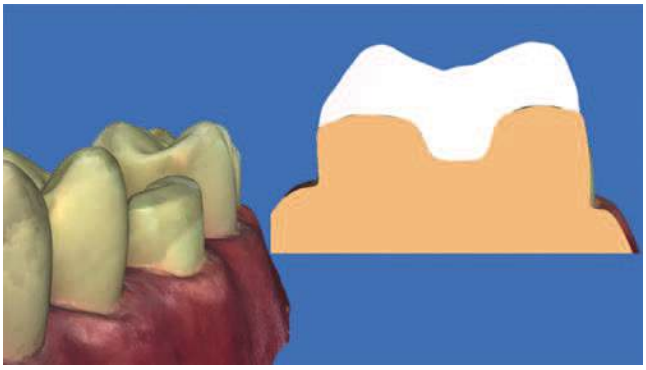


• **Fig. 12.8** Ferrule margin design with opposing walls for mechanical retention more appropriate for cast-metal restorations.



• **Fig. 12.9** Preferred butt-joint margin design for ceramic onlay preparation.

proper seating of the restoration due to undermilling of the relatively parallel walls and/or sharp transitions. Alternatively the system may significantly overmill the intaglio surface of the restoration, thinning the restoration further. Both undermilling and overmilling are detrimental to the final onlay restoration. Consequently, a butt-joint margin is preferred as it allows for proper thickness of the ceramic material at the margin, reducing the incidence of marginal fracture (Fig. 12.9). Unfortunately, this marginal design often causes a visible demarcation between the tooth and the restoration. Hence, when the blend of the restoration and the tooth is



• **Fig. 12.10** Modified butt-joint margin with a football-shaped diamond for improved esthetic blending of the onlay margin.

of esthetic concern, a modification of the facial butt-joint margin is necessary. The cavosurface margin may be modified with a football-shaped diamond at a 45-degree angle, creating opportunity for a transition of ceramic thickness over the underlying enamel while maintaining a bulk of ceramic at the margin for physical strength of the ceramic (Fig. 12.10). An *in vitro* study on ceramic preparations for conservative restoration of endodontically treated teeth reported a significant improvement in the marginal and internal fit of the ceramic restoration with this modified-margin design relative to both a ferrule and 90-degree butt-joint margin.[10]

Chairside CAD/CAM Clinical Workflow

The clinical workflow to design and fabricate a chairside CAD/CAM restoration is relatively similar for all currently available systems, with noticeable differences in the unique cameras, software programs, and milling chambers of marketed systems (Fig. 12.11A–L workflow sequence). The tooth preparation, including the margins, must be visible by retraction of soft tissues and isolated from moisture contamination so that the camera may accurately capture a digital image of the preparation. The opposing arch is also recorded, as well as a scan of the dentition in maximum intercuspation from the facial aspect. The computer software virtually articulates the opposing models using the scan of the facial surfaces so the appropriate occlusal relationships for the restoration may be designed. The restoration design is initiated by identifying the margins of the planned restoration to identify the limits of the restoration design. The software utilizes data from adjacent and opposing teeth to generate a proposed (virtual) restoration, which may then be edited by a series of tools unique to each software program. The design software enables control of the emergence profile of the restoration, occlusal anatomy, planned occlusal contacts, and size and intensity of proximal contacts. Once the design of the restoration is complete, it is transmitted to the milling unit so the fabrication process may create the volumetric shape of the restoration out of an industrially produced block of ceramic material. Although a variety of chairside fabrication processes may be imagined, subtractive grinding or milling processes of ceramic blocks are accomplished based on the volumetric design created with the software design program. Most restorations may be designed and milled within 10 to 15 minutes to allow for delivery of the restoration at the same appointment. Extended machining times may be required for more complex geometries of fixed partial dentures or implant restorations. Slower machining may be preferred for thin margins, such as for porcelain veneers, in order to prevent

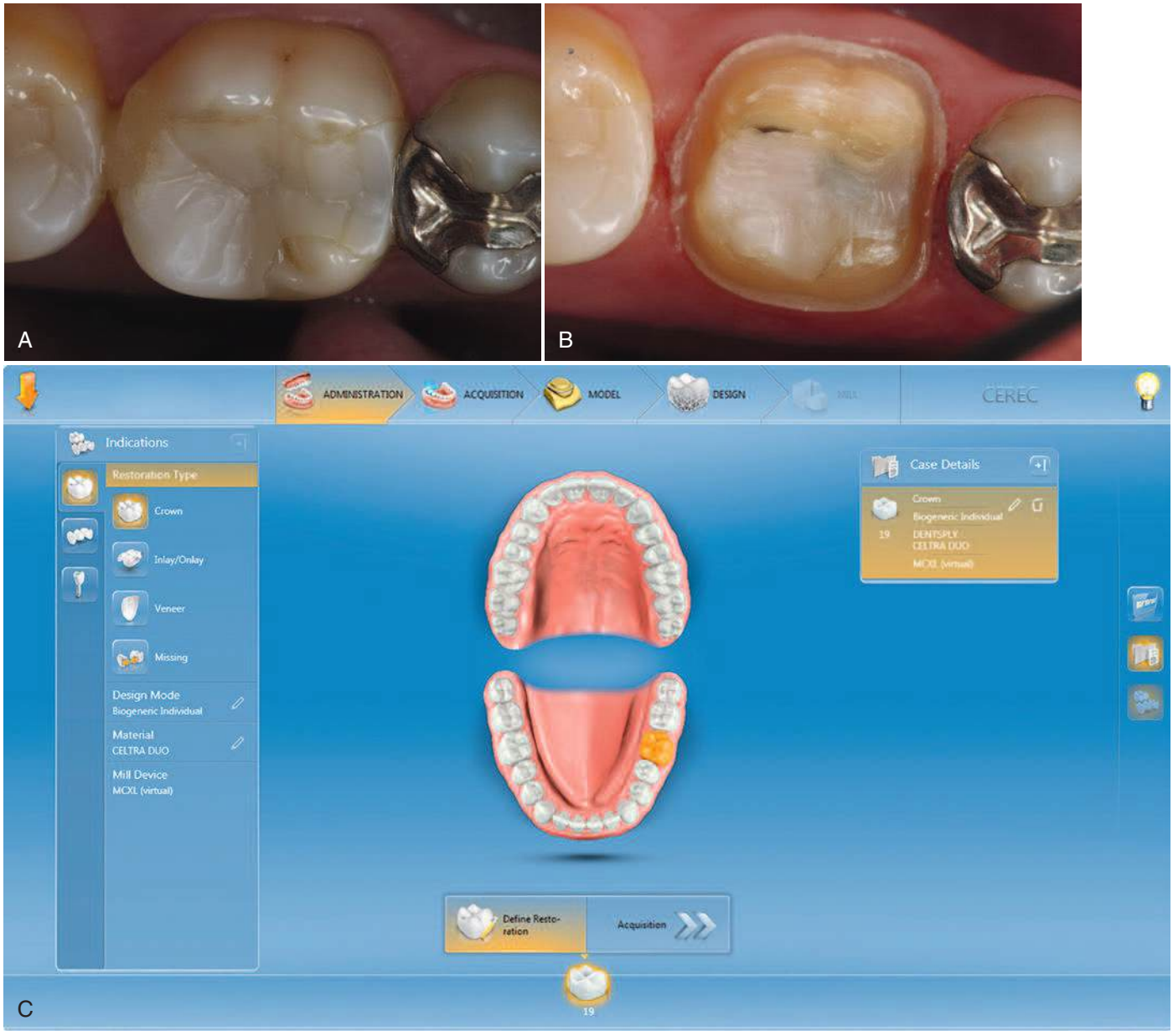
chipping during fabrication. Certain systems allow for fine machining modes, which slow the advancement of the block at each pass of the diamonds/burs (or allow for a smoother second machining pass), so as to refine the surface contour. Final glaze firing and/or external surface polishing follows clinical confirmation of the anatomy and marginal adaptation of the resultant restoration. The final restoration is adhesively cemented with resin cement.

Chairside CAD/CAM software programs also offer various design techniques that provide the opportunity to copy a pretreatment or prototype tooth form onto the recorded tooth preparation. The copy function minimizes the need for restoration editing following the proposal of the restoration, which is especially useful in clinical situations requiring the restoration of fractured abutment teeth for removable partial dentures (RPD). Existing rest seat and proximal contours are copied from the pretreatment tooth and subsequently applied to the recorded tooth preparation. The new ceramic restoration will then have very good adaptation to the existing RPD. The copy-design process is also very useful for anterior restorations through use of a diagnostic wax-up containing strategically planned contours of the restorations. These contours may be virtually copied onto the recorded preparation, digitally perfected, and then transmitted for milling.

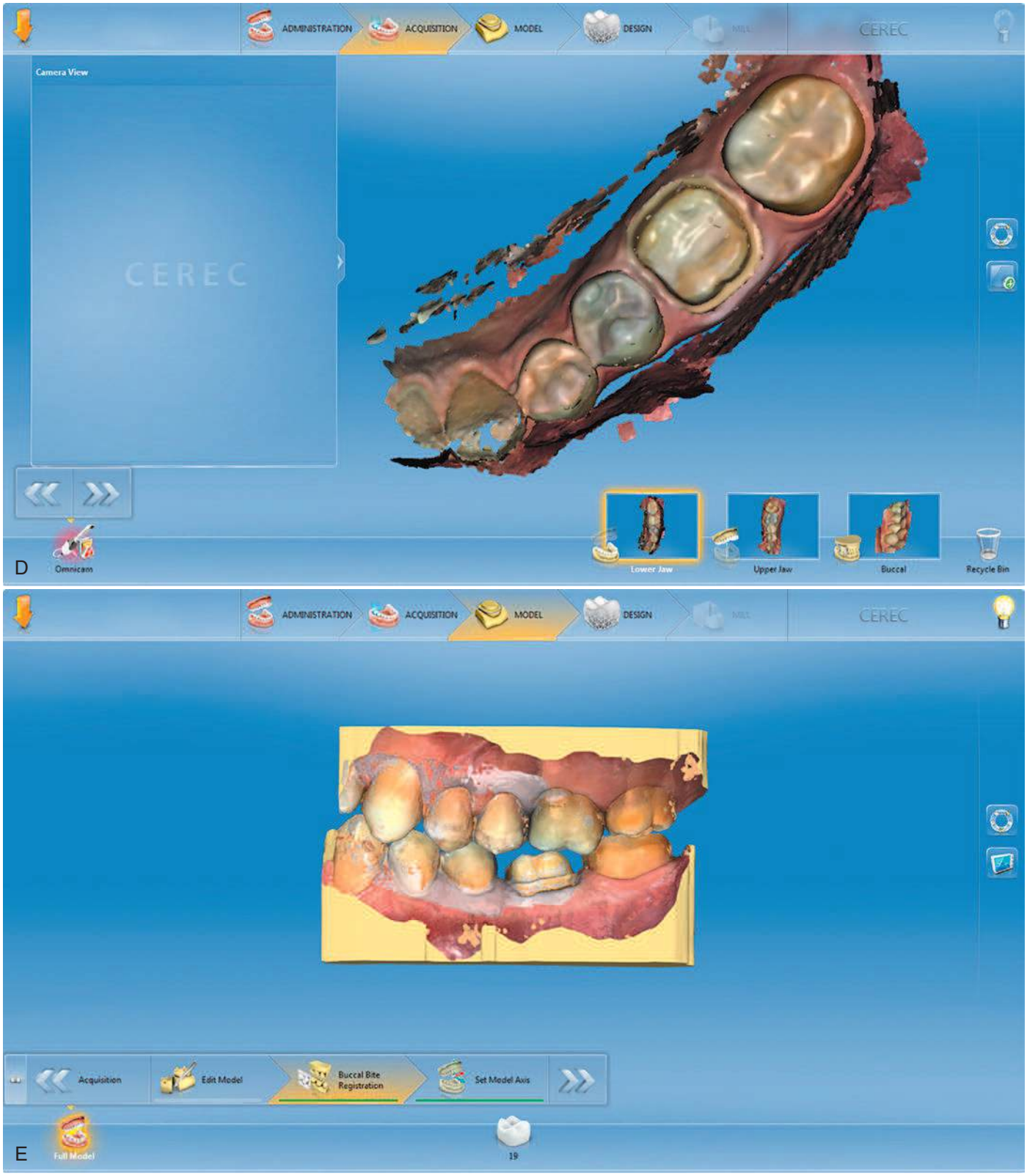
Chairside Restorative Materials

Dental material manufacturers produce monolithic materials for chairside CAD/CAM restorations generally referred to as "blocks." The blocks are dense, homogeneous materials that have been industrially produced under ideal conditions so as to limit the presence of internal flaws (porosity or voids). Limited inclusion of flaws enables maximization of the physical properties of the

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• **Fig. 12.11** Chairside CAD/CAM workflow using the CEREC system. A, Preoperative view of tooth #19. B, All-ceramic crown preparation for tooth #19. C, Case identification in the Administrative Phase.

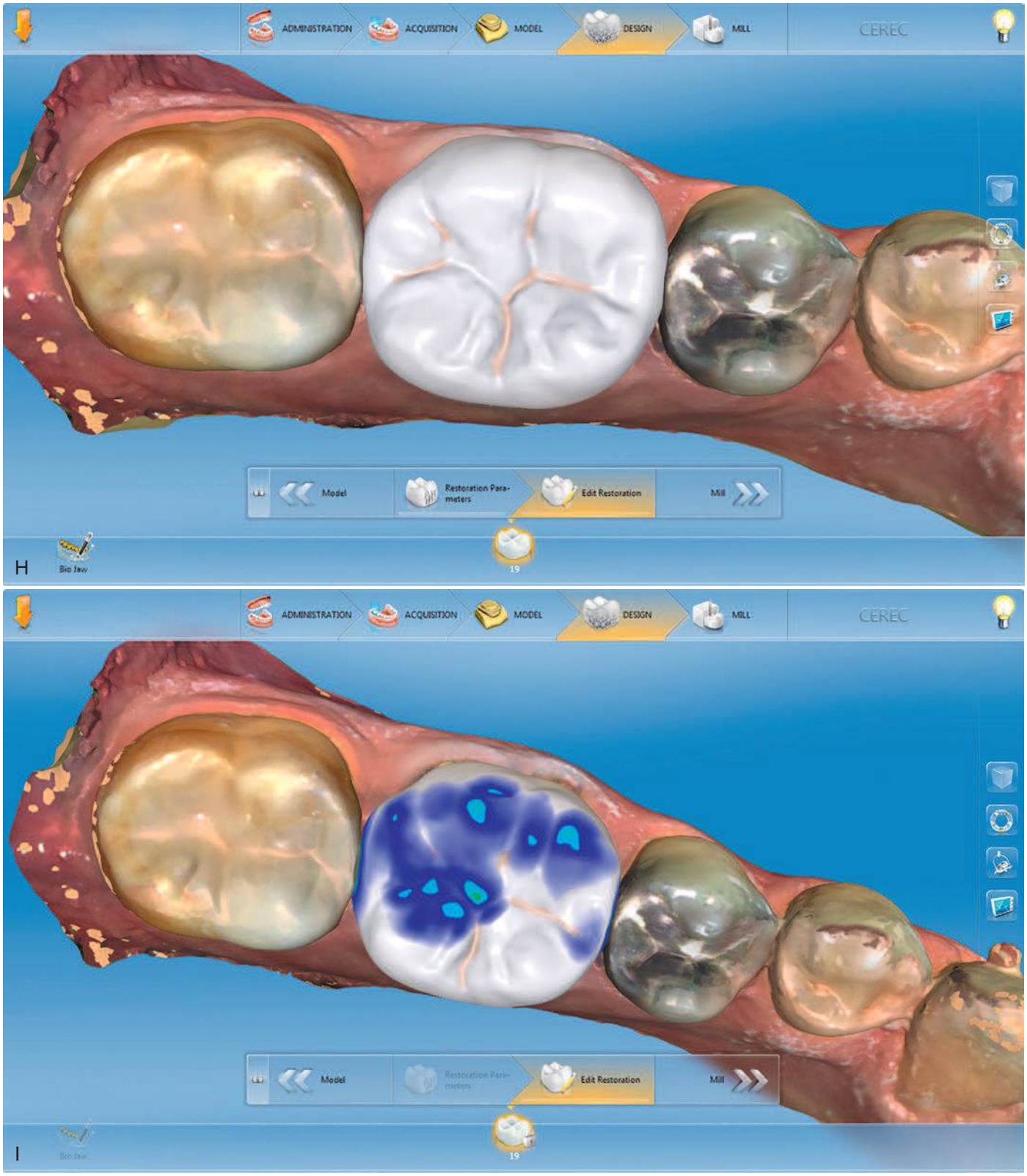


• **Fig. 12.11, cont'd** D, Intraoral digital scan of the mandibular left quadrant. E, Virtual mounting of the opposing digital models.

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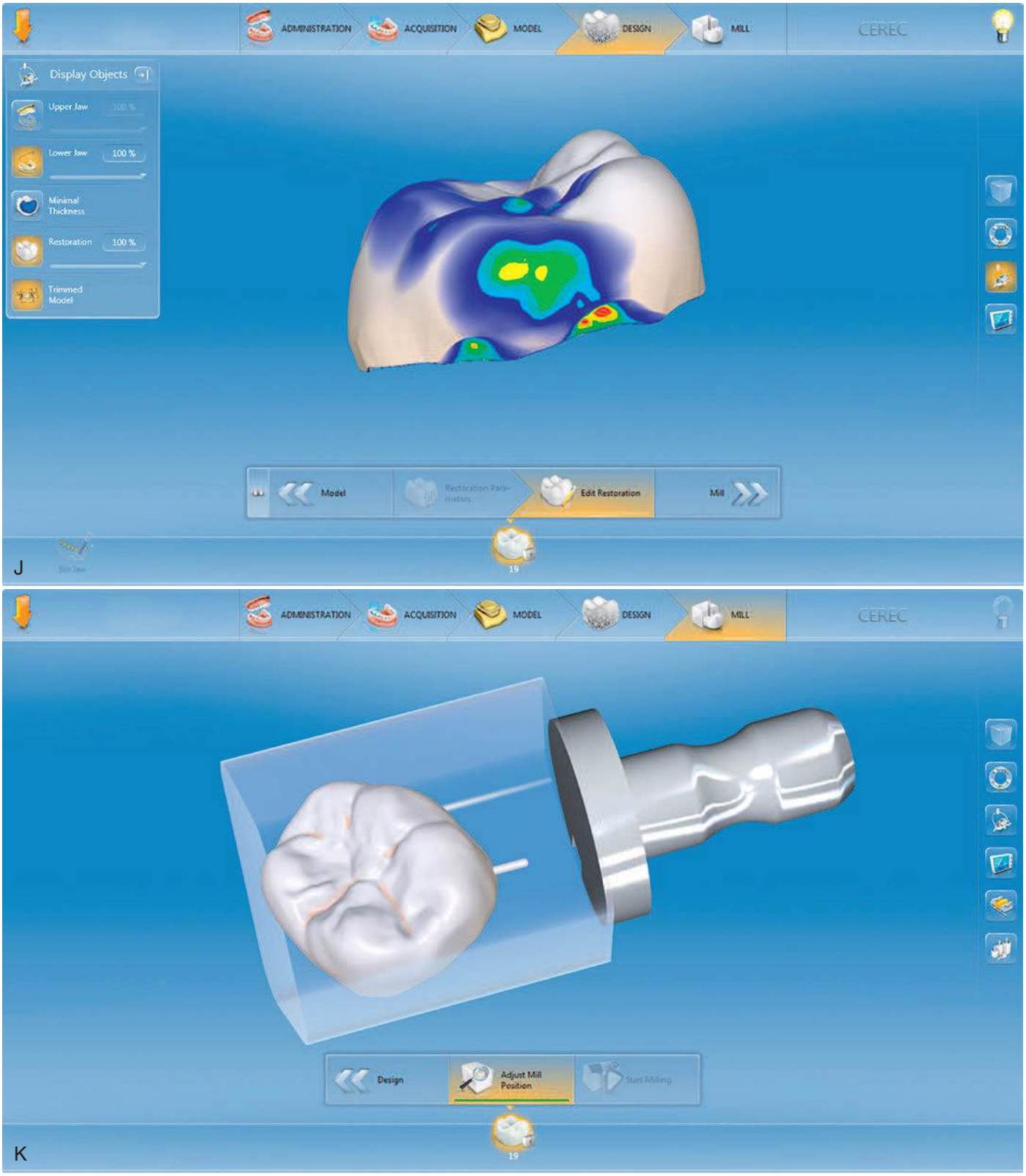


• **Fig. 12.11, cont'd** F, The Model Axis is set for the model. G, The crown margin is identified.



• **Fig. 12.11, cont'd** H, Biogeneric proposal of the crown. I, Visualization of the occlusal contacts for refinement.

Continued



• **Fig. 12.11, cont'd** J, Visualization of the proximal contact for refinement. K, Crown proposal in the proposed mill block (Celtra Duo).



• **Fig. 12.11, cont'd** L, Delivered Celtra Duo ceramic crown for tooth #19. (C–K, Courtesy Dentsply Sirona.)

TABLE 12.2 Restorative Material Options for Chairside CAD/CAM Systems

Material	Brand	CEREC	PlanFit	CS Solution
Adhesive ceramic				
Leucite-reinforced	IPS EmpressCAD (Ivoclar)	X	X	–
Feldspathic	Vita Mark II (Vita)	X	X	X
	Sirona Blocs (Dentsply Sirona)	X	–	–
High-strength ceramic				
Lithium disilicate	IPS emaxCAD (Ivoclar)	X	X	–
Zirconia-reinforced lithium-silicate	Celtra Duo (Dentsply Sirona)	X	–	–
Lithium-silicate	Obsidian (Glidewell)	–	X	–
Resilient ceramic (resin nano ceramic)	Lava Ultimate (3M)	X	X	X
	Enamic (Vita)	X	X	X
	Cerasmart (GC America)	X	–	–
Composite resin	Paradigm MZ100 (3M)	X	–	–
	Brilliant Crios (Coltene)	X	–	–
Zirconia (full contour)	CEREC Zirconia (Dentsply Sirona)	X	–	–
Provisional acrylic materials	TelioCAD (Ivoclar)	X	X	–
	Vita CAD-Temp (Vita)	X	–	–

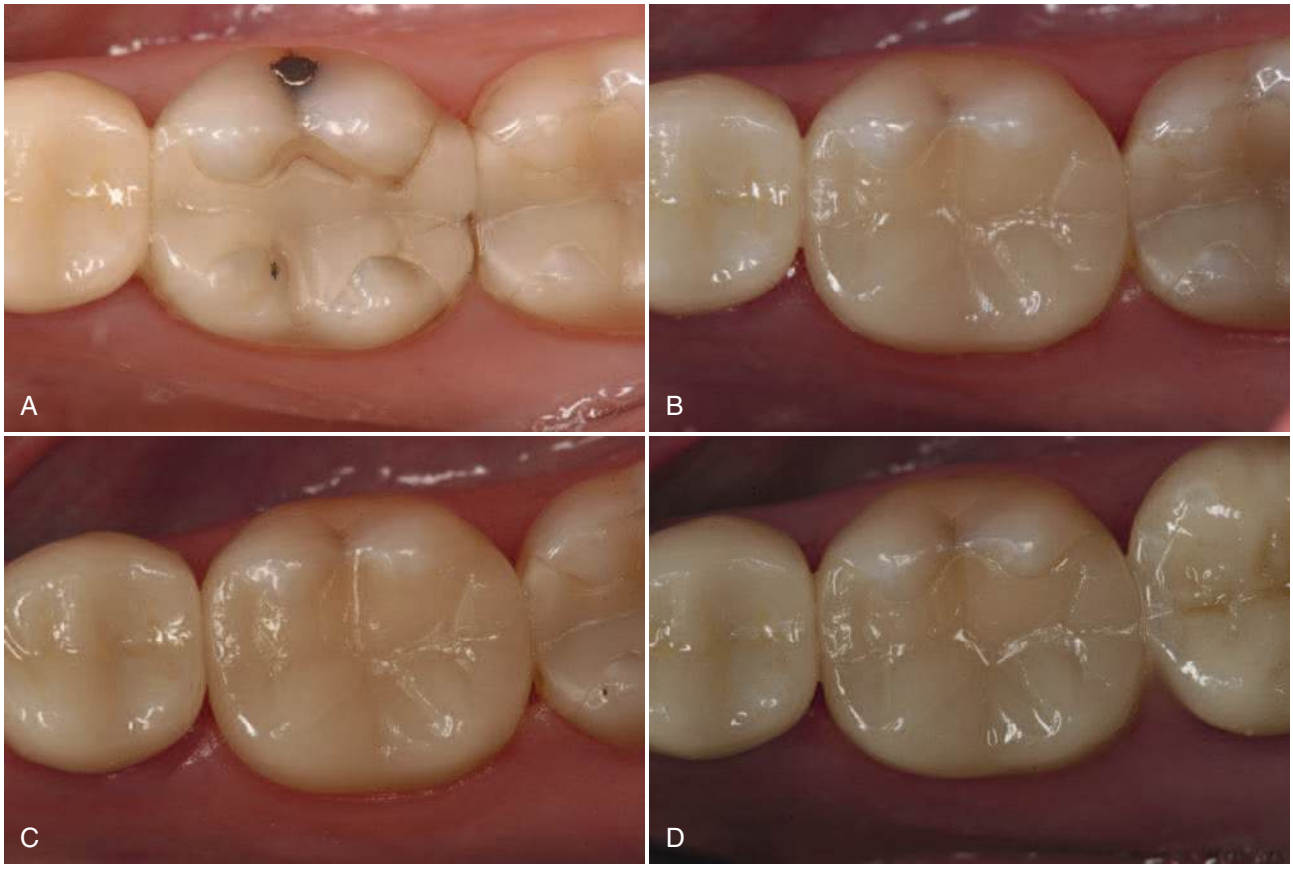
material. The blocks are attached to mandrels that are specific to the various brands of chairside CAD/CAM milling units.

The first CAD/CAM block was created as a result of the collaboration between Dr. Werner Mörmann and the Vita Corporation. The Vita Mark I blocks were originally created out of feldspathic porcelain and eventually evolved into the current generation of feldspathic blocks, Vita Mark II. Since then the CAD/CAM market has expanded to include multiple formulations of ceramic materials.

Chairside CAD/CAM materials may be divided among a number of categories based on material composition for ease in understanding their properties and clinical applications.[11,12] These categories include (Table 12.2):

1. Adhesive ceramics (feldspathic, leucite-reinforced) that must be etched and adhesively bonded to the tooth structure
2. High-strength ceramics (lithium disilicate, zirconia reinforced lithium silicate) with improved strength properties compared to the adhesive ceramic materials
3. Resilient ceramics that do not require a porcelain furnace and must be adhesively bonded to the tooth
4. Composite materials that have a resin matrix
5. Full-contour zirconia that may be cemented to the tooth
6. Provisional materials used for temporary restorations

The adhesive ceramic category includes both fine-grained feldspathic porcelain and leucite-reinforced porcelain. These materials contain a significant glass component resulting in increased translucency allowing them to have an improved "chameleon" effect. The chameleon effect may be described as the ability to reflect the color of the surrounding tooth structure and thus blend into the existing tooth shade (Fig. 12.12). These materials have a moderate flexural strength on the order of 100 to 175 megapascals (MPa). The glass component of these materials allows them to be etched with hydrofluoric acid for micromechanical adhesive bonding. Adhesive bonding is critical to their long-term success because the glass matrix makes them brittle. These materials do not have enough inherent physical strength to be cemented with resin-modified glass ionomer or traditional glass ionomer cements. The adhesive resin cement not only provides retention for the restoration but, by virtue of its adhesive properties, contributes to the physical strength of the ceramic to resist fracture by limiting flexure under



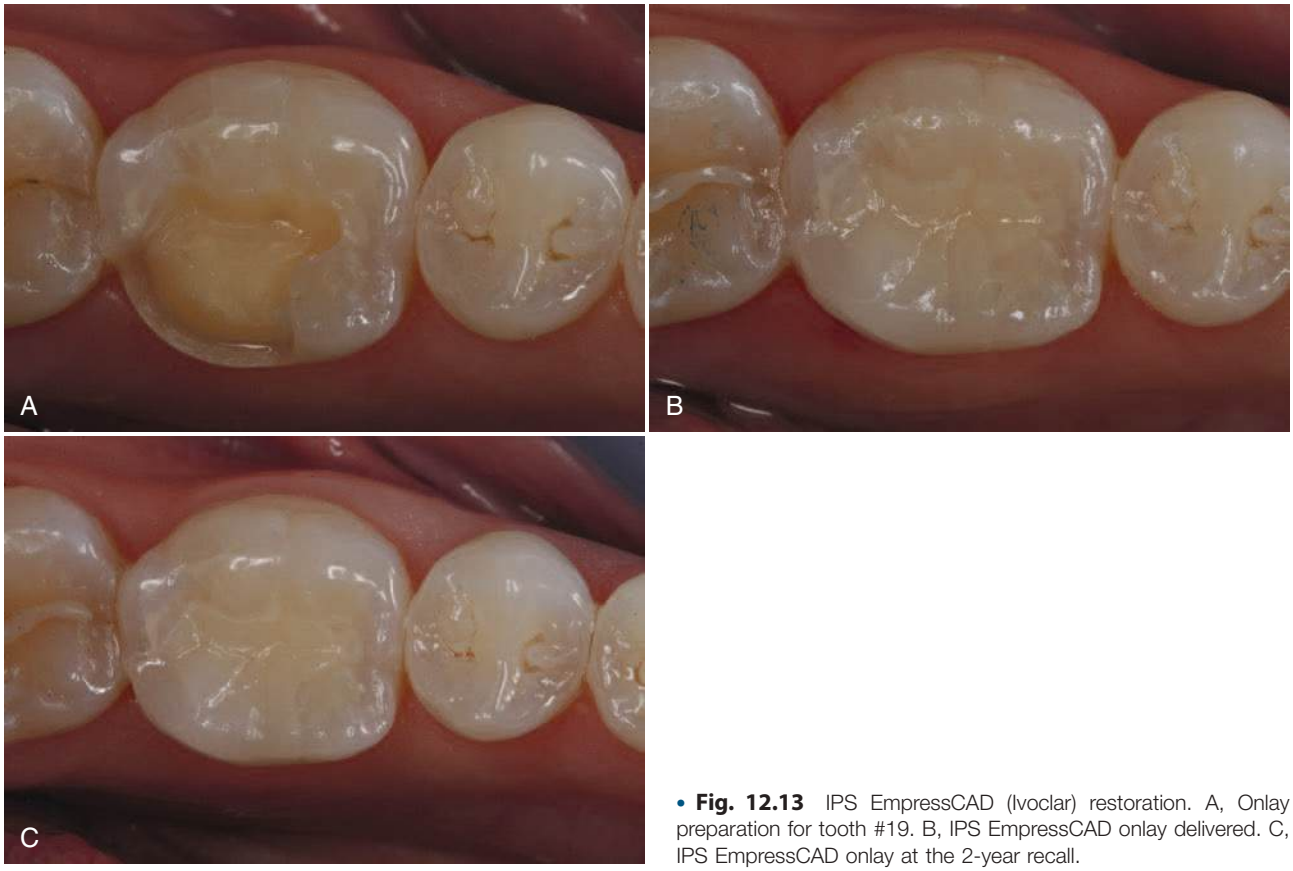
• **Fig. 12.12** Vita Mark II (Vita) onlay for tooth #30. A, Preoperative view of tooth #30. B, Vita Mark II onlay delivered. C, Vita mark II onlay at the 3-year recall. D, Vita mark II onlay at the 5-year recall.

occlusal load. Vita Mark II (Vita) and Sirona Blocks (Dentsply Sirona) are both feldspathic porcelain blocks with a fine-grained particle size that averages 4 microns and a flexural strength of 100 MPa.[13,14]

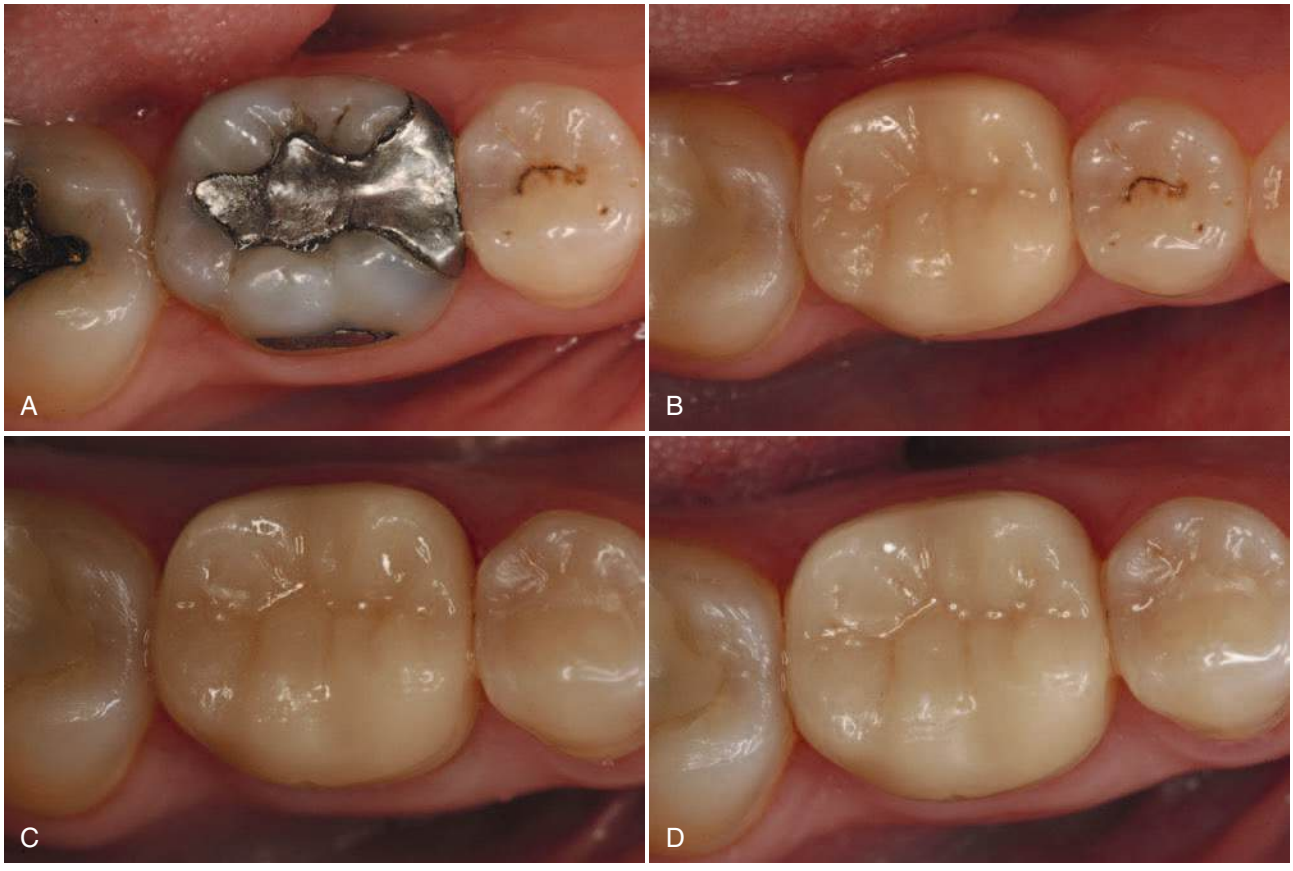
The leucite-reinforced glass ceramic was initially introduced as ProCAD (Ivoclar) and later evolved into IPS EmpressCAD (Ivoclar) with a similar crystal structure as IPS Empress 1 (Ivoclar). EmpressCAD has a finer particle size of 1 to 5 microns that is evenly dispersed 35% to 45% by volume using a proprietary manufacturing process. It comes in two levels of translucency as well as multishaded blocks.[13] The manufacturer-reported flexural strength is 160 MPa, which is within the range of what is reported independently in the literature. EmpressCAD also has a light scattering behavior that allows for a natural chameleon effect (Fig. 12.13).

There are several higher strength glass ceramic materials available for chairside CAD/CAM restorations. IPS e.maxCAD (Ivoclar) is composed of lithium disilicate and has a significantly greater flexural strength and fracture toughness than other adhesive glass ceramics.[15,16] The manufacturer provides the IPS e.maxCAD block in a partially crystallized state (160 MPa, 40% crystalized by volume) to allow for easier and more efficient grinding of the material. After the restoration has been fabricated, it must be subjected to a two-stage firing cycle in a ceramic furnace under vacuum to complete the crystallization process and achieve the maximum flexural strength potential of the material (500 MPa). The resulting glass ceramic restoration has a grain size of approximately 1.5 microns with a 70% crystal volume incorporated in a glass matrix. The crystallization firing also creates optimum translucency for the material (Fig. 12.14). Celtra Duo (Dentsply Sirona) is a zirconia-reinforced lithium silicate (ZLS) material that is another example of ceramic with high strength. The ZLS microstructure has high content of ultrafine glass ceramic crystals (<1 μm) and 10% zirconia content. It is provided by the manufacturer in a fully crystallized state that may be either hand polished or glaze fired in a ceramic furnace prior to delivery. Hand polishing the restoration results in a material that has a flexural strength of 210 MPa, while glazing it in a porcelain oven results in a restoration with a flexural strength of 370 MPa. Celtra Duo is available in shades A1, A2, A3, A3.5, and B2 in both high and low translucencies (Fig. 12.15). NICE! (Straumann) is another fully crystallized zirconia-reinforced lithium silicate block that has recently been introduced for chairside CAD/CAM restorations.

The Resilient Ceramic category of chairside CAD/CAM blocks [Lava Ultimate (3M), Enamic (Vita), and Cerasmart (GC America)] includes materials that have a resin matrix (instead of a glass matrix) that, according to manufacturers, allows greater force absorption capability without fracture. The added feature of these materials is that no additional firing is required allowing for a very efficient delivery process following fabrication of the restoration. All three materials (Lava Ultimate, Cerasmart, and Enamic) exhibit flexural strength properties similar to human dentin, and are indicated for single-unit restorations. In contrast, the low stiffness properties of these materials may be considered disadvantageous from a bonding perspective, as marginal seal debonding due to material flexure might occur.[17] Therefore treating the intaglio surface of the



• **Fig. 12.13** IPS EmpressCAD (Ivoclar) restoration. A, Onlay preparation for tooth #19. B, IPS EmpressCAD onlay delivered. C, IPS EmpressCAD onlay at the 2-year recall.



• **Fig. 12.14** IPS emaxCAD (Ivoclar) crown for tooth #30. A, Preoperative view of tooth #30. B, IPS emaxCAD crown delivered. C, IPS emaxCAD crown at 2-year recall. D, IPS emaxCAD crown at 4-year recall.



• **Fig. 12.15** Glazed Celtra Duo (Dentsply Sirona) crown for teeth #4 and #5. A, All-ceramic crown preparations for teeth #4 and #5. B, Facial view of all-ceramic crown preparations from the facial. C, Delivered glazed Celtra Duo crowns for teeth #4 and #5. D, Facial view of delivered Celtra Duo glazed crowns.

restoration with sandblasting technique as well as selecting a total-etch approach combined with adhesive cementation is considered to be imperative. Because resilient ceramics are less dense they mill faster, with a smaller incidence of margin chipping during milling compared to glass-containing materials.[17]

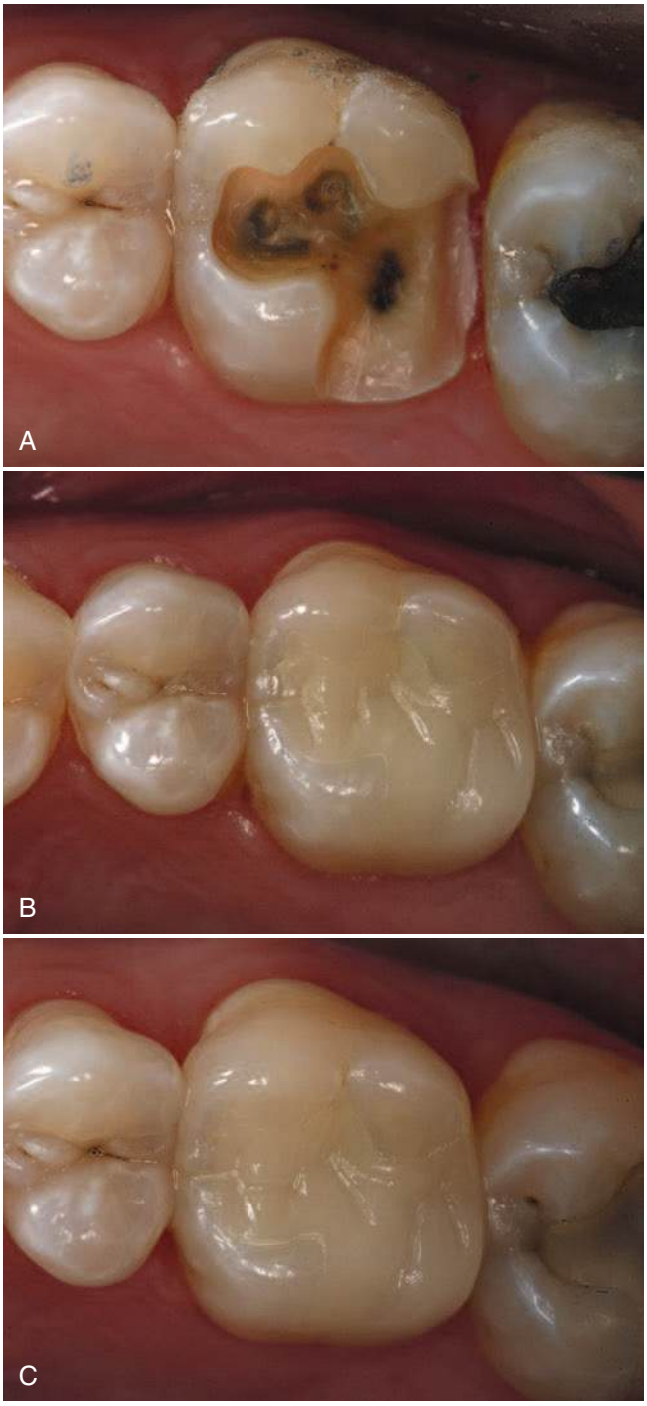
Lava Ultimate (3M) is a nanoceramic material that contains 20 nanometers (nm) size silica particles, 4 to 11 nm size zirconia particles, and agglomerated nanosize particles of silica and zirconia, all embedded in a highly cross-linked polymer matrix with an approximately 80% ceramic load. Lava Ultimate has a reported flexural strength of 170 MPa and it is indicated for inlays and onlays but not for crowns (Fig. 12.16).[17,18]

Cerasmart (GC America) is described by the manufacturer as a flexible nanoceramic with a resin matrix containing homogeneously distributed nanoceramic filler particles. The material is a high-density composite resin with 71% silica and barium glass nanoparticles filler by weight.[17] The reported flexural strength of Cerasmart is 230 MPa and it is indicated for inlays, onlays, and crowns.[17]

Enamic (Vita) is described by the manufacturer as a resin-based (14 wt%) hybrid ceramic comprised of a dual interpenetrating structure of a leucite-based and zirconia-reinforced ceramic network (86 wt%). The mechanical properties of the material are in between glass ceramics and highly filled composites.[19] The ceramic network provides wear resistance; however, it makes the material more brittle and susceptible to fracture. The polymer network is capable of improving the fracture resistance of the material due to its capability of undergoing plastic deformation.[20] The manufacturer recommends the material for inlays, onlays, and crowns and reports that flexural strength of Enamic is 150 MPa, which is significantly lower than Lava Ultimate or Cerasmart. This is in line with the results of an independent study (approximately 135 ± 25 MPa).[17]

Composite resin block materials are gaining in interest as the efficiency of chairside designing and milling a CAD/CAM restoration has improved. The chairside CAD/CAM workflow allows for easier control of proximal contours, contacts, and occlusal relationships for larger, more complex multisurface composite restorations compared to incremental hand placement of direct composite materials. Paradigm MZ100 (3M) is a composite block based on Z100 composite chemistry and relies on a proprietary processing technique to maximize the degree of cross-linking in the bis-GMA polymer-based composite material. It has zirconia-silica filler, which is radiopaque, and is 85% filled by weight with an average particle size of 0.6 micron.[21] The manufacturer reported flexural strength of Paradigm MZ100 is 150 MPa. An independent study reported slightly higher value 157 ± 30 MPa, which was not significantly different than the flexural strength reported for EmpressCAD.[17] Brilliant Crios (Coltene) is a recently introduced reinforced composite block containing amorphous silica and glass ceramic particles in a cross-linked methacrylate matrix.

The most recent material innovation for chairside CAD/CAM restorations was the introduction, in 2016, of CEREC Zirconia



• **Fig. 12.16** Lava Ultimate (3M) onlay for tooth #14. A, Onlay preparation for tooth #14. B, Lava ultimate onlay delivered for tooth #14. C, Lava Ultimate onlay at the 3-year recall.

(Dentsply Sirona). This material is a precolored, full-contour zirconia that may be processed efficiently for a single-appointment delivery. The fast production time is a result of the simultaneous introduction of the SpeedFire (Dentsply Sirona) furnace that is able to sinter the zirconia restorations in under 20 minutes. Zirconia has a flexural strength and fracture toughness that is generally at least three times as great as glass ceramic materials.[22] CEREC zirconia has a manufacturer reported flexural strength of over 900 MPa; therefore it is suitable for crowns as well as short-span fixed partial dentures (FPDs) that may be cemented with traditional cements.

A unique property of zirconia is that it shrinks volumetrically approximately 22% to 24% when sintered. The CEREC software program automatically compensates for the degree of shrinkage in the milled restoration. When the restoration design is ready for milling, the bar code on the zirconia block is input into the software program. The bar code contains the precise amount of expected shrinkage of the particular zirconia block as determined by the manufacturer. The software virtually expands the designed restoration by this degree and an enlarged restoration is milled. The CEREC software automatically programs the SpeedFire furnace to sinter the milled zirconia restoration and firing shrinkage results in a final restoration that accurately fits the preparation. The sintering process takes 10 to 15 minutes for crowns and more than 25 minutes for FPDs. Subsequent glaze firing and surface color characterization is optional. Glazing of the zirconia restoration may be accomplished in the SpeedFire furnace or in a conventional porcelain furnace.

CAD/CAM blocks for provisional restorations are also available for chairside fabrication of long-term provisional crowns and FPDs. The chairside CAD/CAM process avoids the air-inhibited layer present on conventional self-cure or VLC acrylics as well as polymerization shrinkage. Vita CAD-Temp blocks (Vita) are made from a highly cross-linked, microfilled polymer and are available in extended block sizes, including 40 mm and 55 mm lengths, to accommodate multiple unit FPDs. Telio CAD (Ivoclar) is a millable cross-linked polymethyl methacrylate block that is part of the Telio System (a self-curing composite, desensitizer, and cement). The Telio CAD material is also available in 40 mm and 55 mm size blocks.

Accuracy of Digital Impressions

It is axiomatic that the accuracy of the final restoration depends on the accuracy of the recorded dimensions of the tooth preparation. This is true for both conventional and digital impressions. The accuracy of the margin fit and internal adaptation of any restoration is limited by the geometry of the tooth preparation and the limitations of the recording medium. The tooth preparation must be well isolated from moisture contamination and adjacent soft tissues if an accurate digital impression is to be made.

Current technology will not allow the digital capture of the tooth preparation through saliva, blood, or soft tissue. A simple way to think about digital impressions is that the camera can only accurately record what is clearly visible to the eye of the operator. Careful control of the scanning environment ensures an accurate surface reproduction with the digital impression.

Conventional impression materials require lateral and cervical retraction of the soft tissues for an accurate impression. Lateral space must be adequate to ensure sufficient impression material to resist tearing at the margin. Cervical space, generally 1 mm cervical to the margin, is required for accurate assessment of the recorded margin in the impression. Digital impressions have the advantage that soft tissues only need be retracted laterally in order to visualize the margin.

Digital impressions provide excellent immediate feedback relative to the recorded tooth preparation. Digital magnification of the image, in many cases up to 20 times lifesize, facilitates critical evaluation of the tooth preparation while the patient is still in the

chair. Preparation corrections may be accomplished immediately. Additionally, inadequately captured areas may be immediately reimaged without the need to redo the entire impression, as is the case with conventional impression materials.

Research Relative to CAD/CAM Systems

Dentists have ongoing concern that restoration adaptation to tooth preparations allows for optimal clinical performance. The CEREC system was the initial chairside CAD/CAM system introduced to the dental marketplace and has over 30 years of both laboratory and clinical research documenting the relative accuracy of the fit of CEREC chairside CAD/CAM restorations. Digital camera technology has evolved from a single image capture process (RedCam) to the LED-based BlueCam to the present video version (OmniCam). The fit of CEREC restorations have consistently been considered accurate irrespective to the version of the image capturing device used. Cook and Fasbinder compared marginal fit and internal adaptation of crowns fabricated on virtual models made from the CEREC 3 infrared laser camera (RedCam) and CEREC BlueCam.[10] No significant difference was found in the margin fit and internal adaptation of CEREC crowns fabricated with either RedCam or BlueCam. The average marginal gap was 67 ± 18 μm for all groups evaluated.

Marginal adaptation (or margin fit) plays a critical role in long-term clinical performance of CAD/CAM restorations; when the adaptation of the restoration to the preparation is poor it leads to greater discrepancies at the margin. Restoration adaptation has been thoroughly investigated in the literature. Whereas some internal space is necessary to accommodate the resin cement, most authors are in agreement that margin openings under 100 μm would be desirable. A systematic review evaluated the fit and margin adaptation of CAD/CAM restorations.[23] Two electronic databases were searched for articles published between 2000 and 2012 and cross-matched against predetermined inclusion and exclusion criteria. A total of 230 studies were reviewed and 90 were selected for the meta-analysis, including data on 26 different CAD/CAM systems. The authors concluded that restorations fabricated using CAD/CAM systems have better internal fit than restorations fabricated using traditional methods. The marginal gap of crowns, made from a variety of primarily ceramic materials, ranged from 10 to 110 μm, often with gaps less than 80 μm. It was reported that the limitation of CAD/CAM systems is not achieving a precise margin fit, but the reliability of doing so over a large set of restorations.

Margin fit is also an indicator of accuracy of restorations. An in vitro study by Seelbach et al. compared the accuracy of ceramic crowns fabricated from scans using Lava COS, CEREC AC, and iTero digital impressions with two different conventional impression techniques.[24] A stainless steel master model was scanned 10 times each with the Lava COS and iTero (Cadent) systems, and zirconia crowns were made from each digital impression. The master model was scanned 10 times with the CEREC AC (BlueCam) system and 10 EmpressCAD crowns were fabricated. Ten putty-wash impressions using a single step and a two-step technique were made and zirconia crowns were made from each impression. There was no statistically significant difference in the marginal fit of the crowns fabricated using Lava COS (48 ± 25 μm), CEREC AC (30 ± 17 μm), iTero (41 ± 16 μm), and the single-step putty-wash technique (33 ± 19 μm), whereas the mean margin fit for the two-step putty-wash technique (60 ± 30 μm) was significantly higher than that of the other methods ($P < 0.05$). The mean internal fit was 29 ± 7 μm for Lava COS, 88 ± 20 μm for CEREC AC, 50 ± 2 μm for iTero, 36 ± 5 μm for single-step putty-wash technique, and 35 ± 7 μm for two-step putty-wash technique. There was no significant difference in the mean internal fit for all groups ($P > 0.05$) except the CEREC AC ($P < 0.05$).

Another study compared the fit of Lava DVS zirconia crowns fabricated using Lava COS digital impressions to Vita Rapid Layering Technique crowns using digital impressions with the CEREC AC system.[25] Two posterior crowns were fabricated for each of 14 patients using each digital impression technique on the same tooth preparation. The replica technique was used with all crowns to measure the clinical adaptation and margin fit. The study reported that crowns made with the Lava COS system had a statistically significant better mean marginal fit (51 ± 38 μm) when compared to crowns fabricated using the CEREC system (83 ± 51 μm); however, this difference in fit was below the accepted clinical threshold of 100 μm and therefore it may not be clinically relevant.

Lee et al. investigated the margin fit of ceramic crowns from two different digital impression systems and the CEREC chairside CAD/CAM system.[26] Crowns were fabricated from digital impressions using the Lava COS and Cercon systems and from digital impressions using the CEREC BlueCam and compared to PFM crowns made with a conventional impression technique. The mean marginal gaps were 70.5 ± 34.4 μm for the PFM crowns, 87.2 ± 22.8 μm for Lava, 58.5 ± 17.6 μm for Cercon, and 72.3 ± 30.8 μm for CEREC BlueCam. There were no significant differences in the marginal fit among most of the groups except that Cercon crowns presented significantly smaller marginal gaps than Lava crowns ($P < 0.001$). However, once again this difference in fit was below the accepted clinical threshold of 100 μm and therefore may not be clinically relevant.

Another study investigated different chairside CAD/CAM systems relative to margin fit of ceramic crowns.[27] Three lithium disilicate crown fabrication techniques were included: CEREC 3D BlueCam, E4D laser scanner, and a lost-wax, heat-pressed technique (IPS Empress 1). Microcomputed tomography was used to measure margin discrepancies on the crowns. Heat-pressed crowns had a mean vertical misfit of 36.8 ± 13.9 μm and CEREC crowns had mean vertical misfit of 39.2 ± 8.7 μm, both of which were statistically significantly smaller than the mean vertical misfit recorded for the E4D crowns, 66.9 ± 31.9 μm ($P < 0.05$). In addition, the authors suggested that clinically acceptable fit should be equal or less than 75 μm. The percentage of crowns with a vertical misfit under the 75 μm reported was 83.8% for CEREC and heat-pressed technique, whereas only 65% of the E4D crowns had vertical misfit under that value, potentially indicating better fit accuracy for CEREC and heat-pressed technique.

Ender and Mehl evaluated the in vivo precision of impressions using conventional and digital methods.[28] The conventional methods used were metal full-arch tray or triple tray with PVS material. In addition, eight digital impression systems were also evaluated: Lava True Definition Scanner and Lava COS (3M ESPE), iTero (Cadent), Trios (3Shape), Trios Color (3Shape), as well as CEREC BlueCam (Software 4.0), CEREC BlueCam (Software 4.2), and CEREC Omnicam (Dentsply Sirona). Five subjects each received three quadrant impressions with each of the impression methods. The impressions were superimposed within each test impression group, for each patient, using CAD software (Geomagic Qualify 12, 3D Systems). A perfect match (misfit of zero) would indicate ideal precision, with increasing values from zero gradually indicating

poorer precision. The precision for all methods ranged from 18.8 ± 7.1 μm for the metal full-arch tray to 58.5 ± 22.8 μm with the triple tray. The conventional metal full-arch tray had statistically best precision of all groups, followed by True Def (21.7 ± 7.4 μm), Trios (25.7 ± 4.9 μm), Trios Color (26.1 ± 3.8 μm), and group BlueCam 4.0 (34.2 ± 10.5 μm). No statistically significant difference was found between BlueCam 4.2 (43.3 ± 19.6 μm), OmniCam (37.4 ± 8.1 μm), Lava COS (47.7 ± 16.1 μm), iTero (49 ± 12.4 μm), and conventional triple tray impressions (58.5 ± 22.8 μm). The authors concluded that despite significant differences found, all of the digital impression systems were capable of producing quadrant impression with precision that was clinically acceptable.

Hack and Patzalt measured the ability of six intraoral scanners to accurately capture a single molar abutment tooth in vitro.[29] The scanners tested included iTero, True Def, PlanScan, CS 3500, Trios, and CEREC Omnicam. A master typodont model of a single crown preparation was scanned with a highly accurate industrial benchtop scanner, and the digital file from the benchtop scanner was compared to the digital scans of the intraoral scanners using a CAD software program (Geomagic Qualify). Precision was measured by superimposing the benchtop scanner digital file onto the digital files recorded by each scanner and evaluated for three-dimensional deviations. Similar to a previously mentioned study, a perfect match (misfit of zero) would indicate ideal precision, with increasing values from zero gradually indicating poorer precision. The authors reported most accurate precision values for the Trios (4.5 ± 0.9 μm), followed by True Def (6.1 ± 1 μm), iTero (7 ± 1.4 μm), CS3500 (7.2 ± 1.7 μm), CEREC OmniCam (16.2 ± 4 μm), and PlanScan (26.4 ± 5 μm). Statistically significant differences were reported between all scanners and CEREC Omnicam ($P < 0.05$) as well as between all scanners and PlanScan ($P < 0.05$). The authors concluded that, even though there were differences in levels of accuracy, all scanners investigated produced clinically acceptable accuracy.

A more recent systematic review also evaluated the marginal fit of single-unit all-ceramic crowns fabricated after conventional and digital impressions. The search included studies published from January 1989 through December 2014. From the original 63 articles that were identified, 12 were selected (both *in vitro* and *in vivo*) after applying the inclusion and exclusion criteria. For *in vitro* studies mean marginal fit of crowns fabricated after conventional impressions was 58.9 μm (95% CI: 41.1–76.7 μm), whereas for digital impressions it was 63.3 μm (95% CI: 50.5–76 μm). However, for *in vivo* studies the results were reversed with mean marginal fit of crowns fabricated after digital impressions 56.1 μm (95% CI: 46.3–65.8 μm), whereas for digital impressions it was 79.2 μm (95% CI: 59.6–98.9 μm). The authors reported low overall risk of bias of the included studies as well as moderate heterogeneity, mostly due to the variety of intraoral scanners and ceramic materials in the studies included. The result of the meta-analysis indicated no statistically significant difference in marginal gap of single-unit ceramic restorations fabricated after conventional or digital impressions. The authors suggested that the digital workflow exceeds clinically acceptable standards and performs equal to conventional impressions.[30]

The PlanFit and CS Solutions systems have been more recently introduced for chairside CAD/CAM restorations and have very limited published research on margin fit and internal adaptation. One study measured the marginal fit of E4D fabricated crowns on typodont preparations completed by 62 different clinicians.[31] Each of the crown preparations was judged as good, fair, or poor by using both a replica technique and visual examination looking at common criteria for ceramic restorations. The quality of the preparation influenced the accuracy of the margin fit in ways that were statistically significant. Ideal preparations had a mean margin fit of 38.5 ± 9 μm, fair preparations had 58.3 ± 12 μm, and poor preparations had 90.1 ± 23 μm. A second *in vitro* study measured the margin fit and internal adaptation of E4D fabricated emaxCAD crowns.[32] Mean margin fits varied from 79.32 ± 63.18 μm for buccal margins to 50.39 ± 35.98 μm for lingual margins. Additional independent studies are warranted for a more precise assessment of margin fit and internal adaptation of PlanFit and CS Solutions restorations.

Clinical Longevity of CAD/CAM Restorations

The clinical performance of chairside CAD/CAM restorations has been studied so as to assess relative longevity in the oral environment. Long-term randomized clinical trials are considered the most robust study design for the purpose of proper assessment of clinical longevity. However, there is considerable amount of variation in the dental literature when searching for independent randomized studies on clinical longevity of chairside CAD/CAM restorations. The vast majority of the publications report on the long-term clinical performance of CEREC restorations primarily because the CEREC System has been available since the early 1990s whereas other systems have been more recently introduced. Systematic reviews represent invaluable tools to synthesize the results of these multiple and varied clinical studies; however, the conclusions of systematic reviews are only as valuable as the quality of the studies they seek to combine.

Wittneben and coworkers evaluated the clinical performance of CAD/CAM restorations in a systematic review.[33] The authors performed an electronic search of all publications on clinical performance of CAD/CAM restoration between 1985 and 2007. Studies were selected following specific inclusion criteria. The included publications comprised 14 prospective and 2 retrospective studies on the chairside CEREC System (CEREC 1 and 2) as well as the laboratory system Celay, providing follow-up data from 2 to 10 years. The search yielded data on a total of 1957 restorations with a mean exposure time of 7.9 years. The restorations included mostly posterior crowns, but some studies evaluated inlays, onlays, endocrowns, and anterior crowns. A total of 170 failures were reported, at a rate of 1.75% failure per year. The most common modes of failure described were fractures of the restoration or tooth. The estimated survival rate for CEREC single-tooth restorations was 91.6% (95% CI: 1.22%–2.52%). Restorations fabricated with feldspathic porcelain had the highest 5-year survival rate, which contrasted with the lowest 5-year survival rate for glass ceramic. At 5 years, ceramic onlays performed equally as successful as crowns; however, the authors identified a significant lack of randomized studies as well as a shortage of scientific evidence longer than 3 years.

Among the CEREC literature, several of the studies published data on CAD/CAM generated feldspathic porcelain restorations. Posselt and Kerschbaum conducted a retrospective study on the clinical performance of 2328 inlays and onlays for 794 patients in a private practice setting. A total of 35 failures were reported over 9 years. The Kaplan-Meier survival probability reported was 97.4% at 5 years and 95.5% at 9 years for CEREC inlays and onlays. According to the authors, failure rate was not significantly influenced by restoration size, tooth vitality, previous presence of deep caries, type of tooth treated, or whether the restoration was

located in the maxilla or mandible. The most common type of failure reported was tooth extraction.[34] A 10-year prospective clinical trial on feldspathic CEREC 1 inlays and onlays reported a Kaplan-Meier survival probability of 90.4% after 10 years for 200 restorations placed in private practice. The restorations were placed in 108 patients (62 female, 46 male). The mean age of the patients was 37 years (range 17 to 75 years). All patients presented with good dental care and a low risk for caries. The restorations were fabricated chairside using Vita MK I feldspathic ceramic and were delivered with adhesive technique with luting composite resin instead of resin cement. A total of 15 restorations (8%) failed in 10 years, primarily due to either restoration or tooth fracture.[35] In a follow-up report of that study, the authors reported less than 3% drop in survival rate for the same restorations at 17 years, with reported Kaplan-Meier survival probability of 88.7%. Six restorations failed during the 7-year period between evaluations, once again primarily due to either restoration or tooth fracture.[36] Compared to clinical studies using early versions of the CEREC system, these more recent studies reported better survival probabilities.

The clinical survival of CAD/CAM restorations is influenced by the adhesive cementation, which in turn is maximized with the presence of enamel margins. This was well demonstrated by a longitudinal study on clinical performance of nonretentive feldspathic CAD/CAM overlays (occlusal veneers). A single operator placed 310 CEREC overlays paying particular attention to keeping preparation margins within enamel whenever possible. After 8 years of follow-up, 286 paired onlays were available for evaluation and the calculated survival rate was 99.3%. The only two fractures observed in this study were on maxillary premolars of one patient with occlusal parafunction.[37] The clinical success of these restorations was likely due to the predictable bonding that is often achieved whenever the all-ceramic preparation is within enamel. In addition, an experienced operator may have influenced such remarkable performance, as more inexperienced operators did not achieve similar results. For example, a short-term prospective randomized controlled clinical trial looked at the clinical performance of 68 paired feldspathic CEREC onlays (Vitablock Mark II) that were cemented with a self-adhesive cement (RelyX Unicem, 3M ESPE). The influence of selective enamel etching was evaluated so each patient had one onlay cemented with the self-etch technique and the other one after selective etching. Statistically significant changes were observed for marginal adaptation and marginal discoloration between baseline and 2 years but no statistically significant difference in clinical performance was recorded between the two different cementation groups. Two onlays debonded, one from each group; two onlays fractured from the self-etch group. The 2-year failure rate for CEREC feldspathic onlays was 5.1% for the self-etch technique and 1.7% for the selective etching group. The authors concluded that selective enamel etching did not statistically significantly influence the results.[38] However, the failure to find significance could be attributed to the short follow-up time. Long-term clinical trials indicate that failures tend to continue to increase over time.

A long-term randomized clinical trial was designed to access the longitudinal performance of 80 inlays machined from either a composite resin material (Paradigm MZ100) or feldspathic ceramic (Vitablocks Mark II), cemented with a dual-cured resin cement (RelyX ARC) using a total-etch bonding technique. At the 10-year recall 89% of the inlays were available for evaluation. No statistically significant difference in margin adaptation was evident between the two materials; however, a significant decrease in margin adaptation was noted when inlays of both materials were compared to baseline values due to margin wear of the adhesive luting cement over time. Composite inlays had fewer fractures in 10 years, with a calculated survival rate of 95% for Paradigm MZ100 inlays versus 87.5% for Vitablocks Mark II. The calculated annual failure rates were 0.5% for composite inlays and 1.25% for ceramic inlays.[39]

The vast majority of publications report on feldspathic CAD/CAM materials because they were the first blocks available for chairside milling. More recent studies have evaluated newer CAD/CAM materials. A randomized clinical study compared leucite-reinforced (Paradigm C, 3M ESPE) and feldspathic (Vita Mark II, Vita) CEREC onlays cemented with a self-etching, self-adhesive resin cement (RelyX Unicem, 3M ESPE). No statistically significant differences in the clinical performance were noted between the two materials at 3 years.[40] This trend continued to be observed at the 5-year follow-up as no statistically significant difference was noted in margin adaptation between the two materials either at 5 years or when results were compared to baseline values. However, an increased trend for localized crevice formation was noted over time and a statistically significant difference was found between baseline and 5-year data for both materials when more discriminative criteria were applied.[41]

Chairside CAD/CAM systems were initially limited to the fabrication of inlays and onlays since crowns were not possible with the milling instruments first introduced; therefore most of the initial studies focus on chairside all-ceramic inlays and onlays. With the advent of newer materials and evolution of the milling systems, an increased number of publications focus on chairside CAD/CAM crowns. Reich and coworkers reported a short-term clinical study on the performance of chairside CAD/CAM IPS e.maxCAD crowns fabricated with the CEREC system. Forty-one full-contour e.maxCAD crowns were placed in 34 patients and 39 were available for the 24-month recall. Kaplan-Meier analysis revealed a 2-year survival rate of 97.4%. However, the failures observed were not due to crown fracture; one crown exhibited secondary caries and two crowns received root canal treatment.[42] In contrast, a large longitudinal clinical study evaluated the clinical performance of 100 IPS e.maxCAD CEREC crowns at 2, 4, and 5 years.[43-45] All crowns were fabricated and delivered at a single appointment. The first 62 crowns were placed with either a self-etching bonding agent and resin cement or a self-adhesive resin cement. The last group consisted of 38 crowns that were delivered using an experimental self-etching, self-curing cement. No crown failures were reported during the first 24 months, therefore the 2-year survival rate was 100%. A total of five failures were recorded at 5 years, four crowns debonded (three of which had been delivered with the experimental cement that never made it to market), and one crown fractured, for a 95% survival rate at 5 years. This study is currently ongoing and the optimistic trends continue to be observed.

There is considerable published clinical research on chairside CAD/CAM technology that has revealed it to be a valuable tool for the delivery of high-quality restorations delivered during a single appointment. The ability to adhesively bond ceramic restorations at the time of restoration fabrication minimizes the occurrence of postoperative sensitivity while allowing for more conservative tooth preparation as no preparation retention is required to retain a temporary restoration. Research studies have assessed an extensive variety of resin-based, porcelain, glass ceramic, and high-strength ceramic materials that may be used to provide restorative solutions for inlays, onlays, veneers, and full crowns.

Conclusion

The clinical application of CAD/CAM technology has developed through a continuous evolution of the hardware and software. A variety of restorative materials are available to offer alternative solutions for a variety of clinical problems. Laboratory and clinical research has documented the accuracy, longevity, and effectiveness of CAD/CAM fabricated restorations. The efficient treatment workflow provides an opportunity for patient-centered treatment in one appointment. The evidence-based reliability, as well as the perceived value of same-day dentistry, has helped secure a permanent spot for CAD/CAM dentistry in operative dentistry practice.

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